

**Ahmedabad
University**

CSE400 Fundamentals of Probability in Computing

Project Milestone 3

s1_g7_its

ADAPTIVE TRAFFIC SIGNAL CONTROL USING LAS VEGAS ALGORITHM

- **Application Domain :**
 - Intelligent Transport System
- **Why Uncertainty Exists in the System :**
 - Traffic arrival rates vary stochastically over time
 - Vehicle speeds and driver behavior are unpredictable
 - Sensor measurements contain noise and detection errors
 - Incidents (lane blockage, pedestrian surges, emergency vehicles) are random
 - Demand patterns differ across peak/off-peak periods
- **Core Problem :**
 - Signal timing decisions must be made despite uncertain and time-varying traffic conditions.

- **Objective :**

- Minimize average vehicle delay and queue length at an intersection by optimizing signal cycle and phase timings.

- **Deterministic Baseline Strategy :**

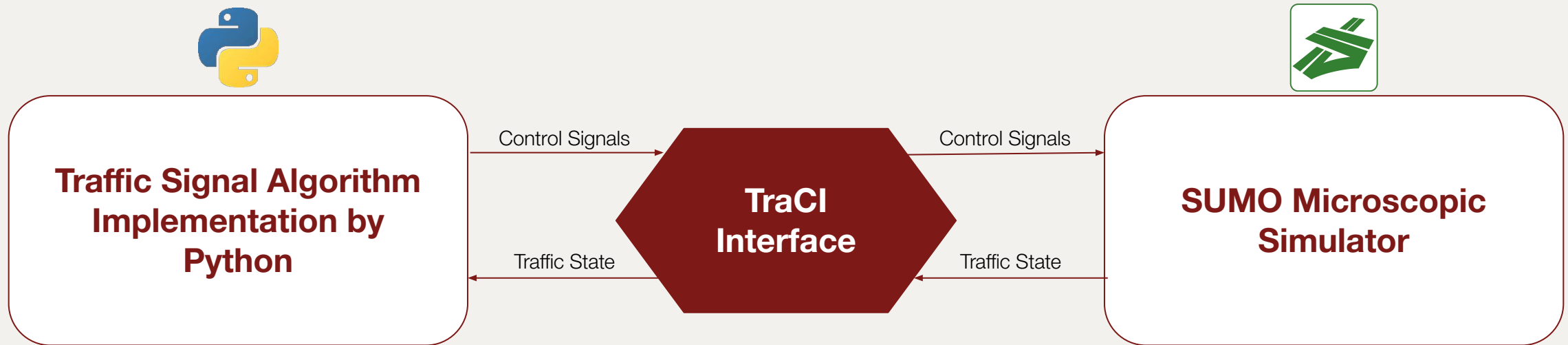
- Use fixed mathematical formulas (Webster / Modified Webster)
- Inputs are treated as exact values (no randomness in decision logic)
- Predefined cycle length and green splits computed analytically
- Same decision rule applied on every simulation run

- **Domain Suitability :**

- Industry-standard classical traffic engineering approach
- Computationally efficient
- Provides a stable reference for performance comparison
- Widely used in real-world fixed-time signal systems

- **End-to-End System Pipeline :**

- SUMO (Simulation of Urban Mobility) as a simulator of the traffic intersection.
- Python is used for the logic implementation and backend control.
- TraCI (Traffic Control Interface) is used as a connection between Python and SUMO.



- **Uncertain Areas :**

- Vehicle arrival generation and turning movement proportions
- speed variations and Flow measurements

- **Noise :**

- Sensor measurement error
- Imperfect queue detection

- **Variability :**

- Random vehicle interarrival times
- Uneven directional demand
- Speed fluctuations

- **Modeling Uncertainty :**

- Simplified flow models
- Saturation flow approximations
- Idealized lane discipline

- **Simulation of Uncertainty :**

- Gaussian noise in queue length
- Randomized vehicle counts
- Stochastic performance modeling
- SUMO microscopic traffic simulation

- **Key Random Variables :**

- Random variables model uncertain traffic states.

- **Primary Random Variables :**

- Queue Length ~ Normal Distribution
- Vehicle count per cycle ~ Normal Distribution
- Passing Rate ~ Normal Distribution
- Waiting Time
- Travel Time
- Vehicle Speed

- **Role in simulation :**

- Determine traffic intensity
- Influence flow ratios
- Affect optimal cycle computation
- Produce stochastic performance outcomes

- **Modeling Principle :**

- Traffic environment → stochastic
- Signal control policy → deterministic

- **Interaction Mechanism :**

- Random Traffic State
 - Deterministic Optimization
 - Deterministic Green Allocation
 - Rule-Based Fuzzy Adjustment
 - Stochastic Performance

Critical Lane Flow Ratio:

$$y_i = \frac{\max(\epsilon_s, \epsilon_l, \epsilon_{sr})}{s}$$

Webster Optimal Cycle:

$$C_o = \frac{1.5L + 5}{1 - \sum y_i}$$

Modified Webster Cycle:

$$C_o = \frac{1.978L + 5.109}{1 - 0.9013 \sum y_i}$$

Green-Time Distribution:

$$G_i = \frac{w_i}{\sum w_i} (C_o - L - mg_{\min}) + g_{\min}$$

- **Deterministic Baseline Logic :**
 - Cycle timing determined analytically each cycle.
- **Operations :**
 - a. Collect vehicle counts from sensors
 - b. Convert counts → hourly flow rates
 - c. Compute critical lane flow ratios
 - d. Compute total saturation $\sum y_i$
 - e. Calculate optimal cycle length
 - f. Allocate green time to each phase
 - g. Execute signal cycle
 - h. Repeat next cycle
- **Key Property :**
 - No randomized control decisions.

- **Simulation Implementation :**
 - Code + SUMO

```
def critical_lane_flow_ratio(eps_s: float, eps_l: float,
                            eps_sr: float, sat: float) -> float:
    """Eq. 1 - highest lane flow ratio across three movement types."""
    return max(eps_s, eps_l, eps_sr) / sat if sat > 0 else 0.0

def webster_cycle(y_sum: float, L: float) -> float:
    """Eq. 3: Co = (1.5L + 5) / (1 - sum_yi)"""
    denom = 1.0 - y_sum
    return (1.5 * L + 5.0) / denom if denom > 1e-6 else float("inf")

def modified_webster_cycle(y_sum: float, L: float) -> float:
    """Eq. 4: Co = (1.978L + 5.109) / (1 - 0.9013 * sum_yi)"""
    denom = 1.0 - 0.9013 * y_sum
    return (1.978 * L + 5.109) / denom if denom > 1e-6 else float("inf")

def compute_optimal_cycle(y_sum: float, L: float, formula: FormulaType,
                          min_c: float, max_c: float) -> float:
    Co_raw = (webster_cycle(y_sum, L) if formula == FormulaType.WEBSTER
              else modified_webster_cycle(y_sum, L))
    return float(np.clip(Co_raw, min_c, max_c))

def distribute_green_times(yi_list: List[float], Co: float,
                           L: float, min_green: float) -> List[float]:
    """Eqs. 9-12: split effective green proportionally to critical flow ratios."""
    m = len(yi_list)
    Ce = max(Co - L - m * min_green, 0.0)
    y_sum = sum(yi_list)
    greens = []
    for yi in yi_list:
        alpha = yi / y_sum if y_sum > 1e-9 else 1.0 / m
        greens.append(round(alpha * Ce + min_green, 2))
    return greens

def flow_to_hourly(n_veh: int, cycle_time: float) -> float:
    """Eq. 6: vehicle count in one cycle -> hourly flow rate."""
    return n_veh * 3600.0 / cycle_time if cycle_time > 0 else 0.0
```



- **Simulation Implementation :**
 - Code + SUMO

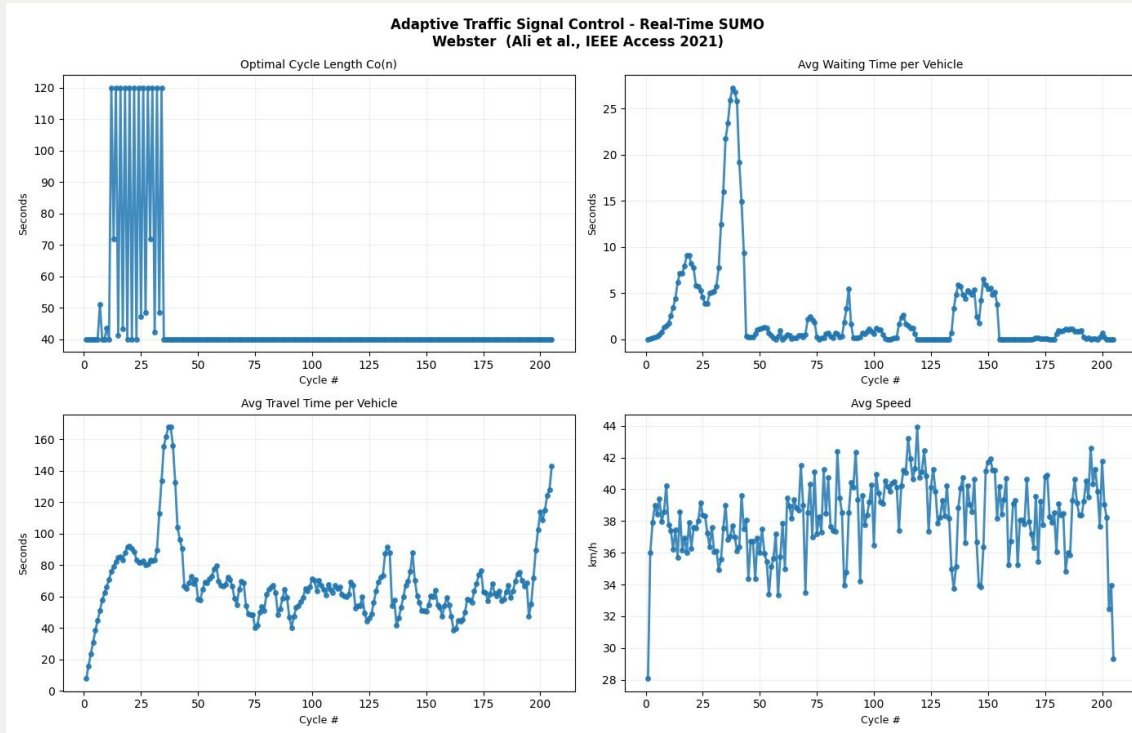


Fig 1: Performance metrics for Webster-based Adaptive traffic control system that ran for about 200 cycles in SUMO. Avg. Waiting time peaks around 30 cycles, and cycle length converging to a minimum around 35 seconds.

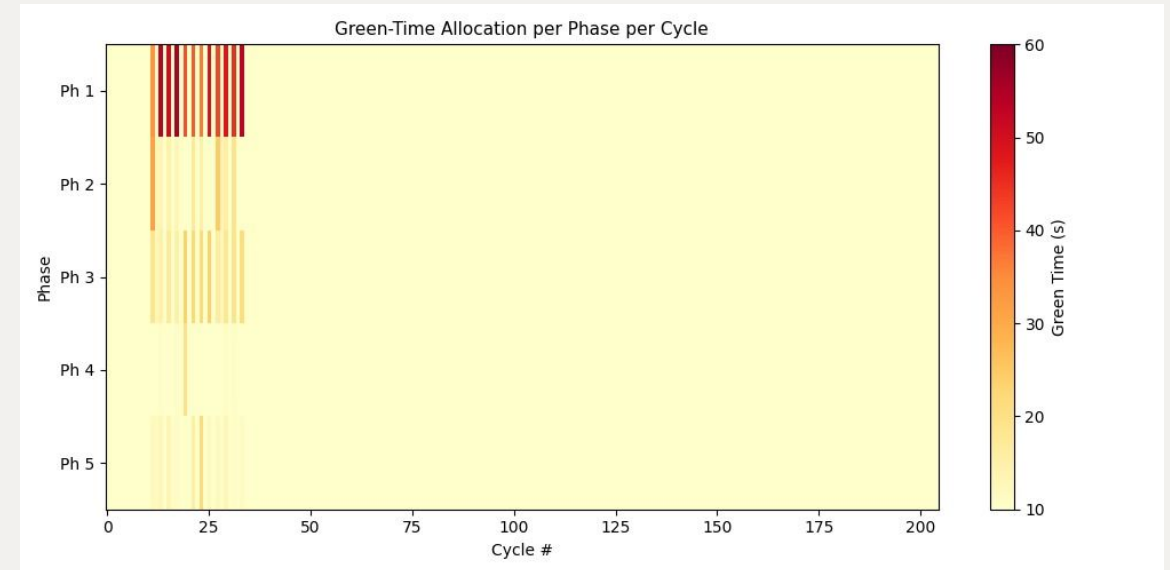


Fig 2: Green time allocation in a 5 signal phase per cycle via Webster's algorithm. Dramatic variations are observed in the initial cycles (0-40), after this all phases stabilize to become the minimum green-time.

- **Limitations of deterministic algorithm :**
 - **Weaknesses :**
 - Depends on accurate flow estimation
 - Formula structure fixed
 - Limited predictive capability
 - Cannot learn long-term patterns
 - **Poor Performance Scenarios:**
 - Highly non-stationary traffic
 - Incident-driven congestion
 - Sensor failures
 - Extreme demand imbalance

- **Key Insights :**

- Deterministic optimization is stable and interpretable
- Fuzzy logic improves short-term responsiveness
- System performs well under moderate variability

- **Future Directions :**

- Reinforcement Learning signal policies
- Probabilistic cycle optimization
- Multi-intersection coordination
- Deep traffic state prediction

Thank You

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M. E. M. Ali, A. Durdu, S. A. Celtek and A. Yilmaz, "An Adaptive Method for Traffic Signal Control Based on Fuzzy Logic With Webster and Modified Webster Formula Using SUMO Traffic Simulator," in IEEE Access, vol. 9, pp. 102985-102997, 2021, doi: 10.1109/ACCESS.2021.3094270.